WORKED SOLUTION

STEP I 2005, Question 1

Pure • Counting Digit Sums

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Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/05-s1-q1

The key idea

The largest possible digit sum of a five-digit number is $9 \times 5 = 45$, and both target sums are close to it. So instead of counting the digits themselves, count how far each digit falls *short* of 9. The shortfalls are small non-negative numbers with a small fixed total, and counting those is easy, either by listing cases or by the "stars and bars" trick.

Solution

Write the number as $d_1d_2d_3d_4d_5$, where $d_1 \in \{1, \ldots, 9\}$ (no leading zero) and $d_2, \ldots, d_5 \in \{0, \ldots, 9\}$. Put

$$e_i = 9 - d_i \qquad (i = 1, \ldots, 5),$$

the *shortfall* of the ith digit. Each e_i is a whole number with $0 \leq e_i \leq 9$ (and $e_1 \leq 8$, since $d_1 \geq 1$), and

$$d_1 + d_2 + d_3 + d_4 + d_5 = 45 - (e_1 + e_2 + e_3 + e_4 + e_5).$$

Different numbers give different lists $(e_1, \ldots, e_5)$ and vice versa, so counting numbers with digit sum S is the same as counting ordered lists of shortfalls with total $45 - S$.

Part (i)

Digit sum 43 means the shortfalls total $45 - 43 = 2$. There are only two ways to split 2 into non-negative parts:

- **One digit is short by 2, the rest by 0.** One digit is 7 and the other four are 9. The 7 can be in any of the 5 positions: 5 numbers $(79999, 97999, \ldots, 99997)$.
- **Two digits are short by 1 each, the rest by 0.** Two digits are 8 and the other three are 9. We choose which 2 of the 5 positions hold the 8s: $\binom{5}{2} = 10$ numbers $(88999, 89899, \ldots, 99988)$.

Every digit is at least 7, so the leading digit is never 0 and all of these really are five-digit numbers. The two cases don't overlap and cover every possibility, so there are

$$5 + 10 = 15$$

five-digit numbers whose digits sum to 43, as required.

Part (ii)

Now the shortfalls must total $45 - 39 = 6$. Since $6 \leq 8$, no single shortfall can exceed its limit: $e_1 \leq 6 \leq 8$ keeps $d_1 \geq 3$, and $e_i \leq 6 \leq 9$ keeps the other digits between 3 and 9. So we simply need the number of ordered solutions of

$$e_1 + e_2 + e_3 + e_4 + e_5 = 6, \qquad e_i \geq 0 \text{ whole numbers.}$$

? **Ask yourself** How can we count these without listing them all?

Picture 6 stars in a row, separated into 5 groups by 4 bars:

$$\star\star \,|\, |\, \star \,|\, \star\star\star \,| \quad \text{stands for} \quad (e_1, e_2, e_3, e_4, e_5) = (2, 0, 1, 3, 0).$$

Every arrangement of 6 stars and 4 bars gives exactly one solution, and every solution comes from exactly one arrangement. An arrangement is fixed by choosing which 4 of the 10 places hold bars, so the number of solutions is

$$\binom{10}{4} = 210.$$

Hence there are **210** five-digit numbers whose digits sum to 39.

If you'd rather count by cases, as in part (i), list the ways to split 6 into at most five positive parts and count arrangements of each:

shortfalls	numbers	shortfalls	numbers
6	5	3, 2, 1	$5 \cdot 4 \cdot 3 = 60$
5, 1	$5 \cdot 4 = 20$	3, 1, 1, 1	$5 \cdot 4 = 20$
4, 2	$5 \cdot 4 = 20$	2, 2, 2	$\binom{5}{2} = 10$
4, 1, 1	$5 \cdot \binom{4}{2} = 30$	2, 2, 1, 1	$\binom{5}{2}\binom{3}{2} = 30$
3, 3	$\binom{5}{2} = 10$	2, 1, 1, 1, 1	5

The total is $5 + 20 + 20 + 30 + 10 + 60 + 20 + 10 + 30 + 5 = 210$, agreeing with stars and bars. (The partition 1, 1, 1, 1, 1, 1 of 6 needs six digits, so it doesn't occur.)

Checks

- Stars and bars reproduces part (i): shortfalls totalling 2 give $\binom{6}{4} = 15$.

- The extreme cases work too: total shortfall 0 gives $\binom{4}{4} = 1$ number (99999), and total shortfall 1 gives $\binom{5}{4} = 5$.
- A computer search through all 90 000 five-digit numbers finds exactly 15 with digit sum 43 and 210 with digit sum 39, with the case breakdown in the table above.

Common slips

- Counting $\{8, 8, 9, 9, 9\}$ as one number, or as 5: the order of the digits matters, and there are $\binom{5}{2} = 10$ arrangements.
- Forgetting to check the leading digit. Here it can't be 0 because every digit is at least 3; for a smaller digit sum (say 20) stars and bars would overcount and the cases with $d_1 = 0$, or digits above 9, would have to be removed.

Beyond STEP

The number of ways to write n as an ordered sum of k non-negative whole numbers is $\binom{n+k-1}{k-1}$, and it is the coefficient of x^n in $(1 + x + x^2 + \cdots)^k$. Restricting each digit to at most 9 replaces this with $(1 + x + \cdots + x^9)^k$, and the cut-off is handled by the inclusion–exclusion principle. These generating-function methods are a staple of first-year combinatorics; Siklos, *Advanced Problems in Mathematics*, and Davenport, *The Higher Arithmetic*, are good places to keep exploring digit and counting problems.

Beyond STEP worked solutions: https://beyondstep.keithmcnulty.org/. Solution text CC0 1.0; the question itself is © Cambridge University Press & Assessment and is not reproduced here.